

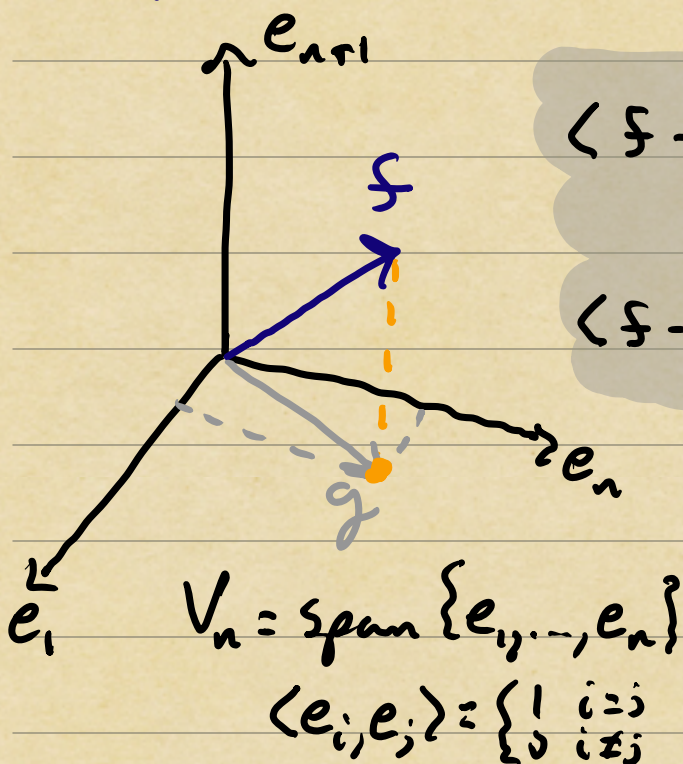
Orthogonal Projection: Orthonormal Bases

Motivation: Given n -dimensional subspace $V_n \subseteq H = \text{Hilbert space}$ and $f \in H$, find best approximation

$$f \approx c_1 e_1 + c_2 e_2 + \dots + c_n e_n$$

$$g_* = \operatorname{argmin}_{g \in V_n} \|f - g\|.$$

Key Idea: Choose g_* s.t. $f - g_* \perp V_n$.



$$\langle f - g_*, e_1 \rangle = 0 \Rightarrow \langle g_*, e_1 \rangle = \langle f, e_1 \rangle$$

$\vdots$

$$\langle f - g_*, e_n \rangle = 0 \Rightarrow \langle g_*, e_n \rangle = \langle f, e_n \rangle$$

$$g_* = \sum_{i=1}^n \langle f, e_i \rangle e_i$$

"orthogonal projection"

Today: Extend this "picture" to separable infinite-dimensional Hilbert spaces.

Orthonormal Bases in infinite-dimensions

Definition: A countable set $\{e_j\}_{j=1}^{\infty} \subset H$ is orthonormal if $\langle e_j, e_k \rangle = \begin{cases} 1 & j=k \\ 0 & j \neq k \end{cases}$. It is an orthonormal basis for H if its span is dense.

$\Rightarrow$ If $f \in H$ and $\varepsilon > 0$, then there is $c_1, \dots, c_n$

$$\|f - \sum_{j=1}^n c_j e_j\| < \varepsilon.$$

Prop 1: Orthonormal set $\{e_j\}_{j=1}^{\infty}$ and let $E_n = \text{span}\{e_1, \dots, e_n\}$. Given $f \in H$, let

$$P_n f = \sum_{j=1}^n \langle f, e_j \rangle e_j.$$

hypotheses

Then $f - P_n f \perp E_n$ and $P_n f$ is the unique minimizer of $\|f - g\|$ over $g \in E_n$.

conclusion

PF To show $f - P_n f \perp E_n = \text{span}\{e_1, \dots, e_n\}$, enough to show that $\langle f - P_n f, e_j \rangle = 0$, $j=1, \dots, n$

$$\begin{aligned} w \in E_n &\Rightarrow \langle f - P_n f, w \rangle = \langle f - P_n f, c_1 e_1 + \dots + c_n e_n \rangle \\ &= \overline{c_1} \langle f - P_n f, e_1 \rangle + \dots + \overline{c_n} \langle f - P_n f, e_n \rangle = 0 \end{aligned}$$

Given e_j , $1 \leq j \leq n$, then

$$\begin{aligned}\langle f - P_n f, e_j \rangle &= \langle f - \sum_{k=1}^n \langle f, e_k \rangle e_k, e_j \rangle \\ &= \langle f, e_j \rangle - \sum_{k=1}^n \langle f, e_k \rangle \langle e_k, e_j \rangle \\ &= \langle f, e_j \rangle - \langle f, e_j \rangle = 0\end{aligned}$$

To show $P_n f$ is unique minimizer of $\|f - g\|$, $g \in E_n$.

$$f - g = f + P_n f - P_n f - g = f - P_n f + P_n f - g$$

$$\|f - g\|^2 = \|f - P_n f\|^2 + \|P_n f - g\|^2$$

$$\Rightarrow \|f - P_n f\|^2 \leq \|f - g\|^2 \text{ for any } g \in E_n$$

" $\Rightarrow$ " holds IFF $\|P_n f - g\| = 0$, therefore IFF $P_n f = g$.

Proposition 1: $\{e_j\}_{j=1}^\infty = \text{ONB}$ for H , then each $f \in H$ has a unique (norm-convergent) rep.

$$f = \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j \quad \left(\|f - \sum_{j=1}^n \langle f, e_j \rangle e_j\| \rightarrow 0 \text{ as } n \rightarrow \infty \right).$$

Equivalently, if $f = \sum_{j=1}^{\infty} a_j e_j$, then $a_j = \langle f, e_j \rangle$ $j=1,2,\dots$

Warning! $f(x) \neq \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j(x)$

Pf Fix $\varepsilon > 0$, then there is $g = \sum_{j=1}^n c_j e_j$ s.t.

(Density) $\|f - g\| \leq \varepsilon$.

From Prop 1, $\|f - P_n f\| \leq \|f - g\| \leq \varepsilon$.

$\Rightarrow \|f - P_n f\| \rightarrow 0$ as $n \rightarrow \infty$.

For uniqueness, suppose $h_n = \sum_{j=1}^n a_j e_j$ converges in norm to f . (Show that $a_j = \langle f, e_j \rangle$)

$$\begin{aligned} |\langle f, e_j \rangle - a_j| &= |\langle f, e_j \rangle - \langle h, e_j \rangle| \\ &= |\langle f - h, e_j \rangle| \\ a_j = \langle h, e_j \rangle &= \langle \sum_{k=1}^n a_k e_k, e_j \rangle \checkmark \end{aligned}$$

$$|\langle f - h_n, e_j \rangle| \leq \|f - h_n\| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\Rightarrow \langle f, e_j \rangle = a_j$$